

Physiology: Solute Transport & Fluid Movement Through Biological Barriers

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Session Objectives

At the conclusion of the session, the student should be able:

- 1. Define and describe the components and underlying physiological forces responsible for transport of H₂O across the plasma membrane**
- 2. Define and describe the roles of intracellular proteins and the Na/K ATPase in regulating the movement of H₂O across the plasma membrane**
- 3. Identify the structural differences in different capillaries and explain their significance in the functions of capillaries.**
- 4. Define and describe the components and underlying physiological forces responsible for transport of H₂O and solutes across epithelial barriers.**
- 5. Apply the principles of H₂O and solute movements in analyzing and understanding the effects of changes in salinity to body fluid compartments.**

You've just spent your last "free" summer trying to get back into shape, as your diet and exercise habits in your 1st year of medical school were horrible. It has been a very hot summer with multiple heat waves.

One very hot afternoon (Heat Index of 103.5°), you decide to go jogging after a session practicing hot yoga. When you leave the studio, you notice that your clothes are very loose and that the veins on your arms and the back of your hands are atypically prominent. You are also very thirsty.



As someone who's going to start hospital training in about a year, you start wondering and ask yourself, "What is going on with me?" and "How can explain this to someone?" and what else is happening that you can't readily observe.

**Vigorous
Exercise**



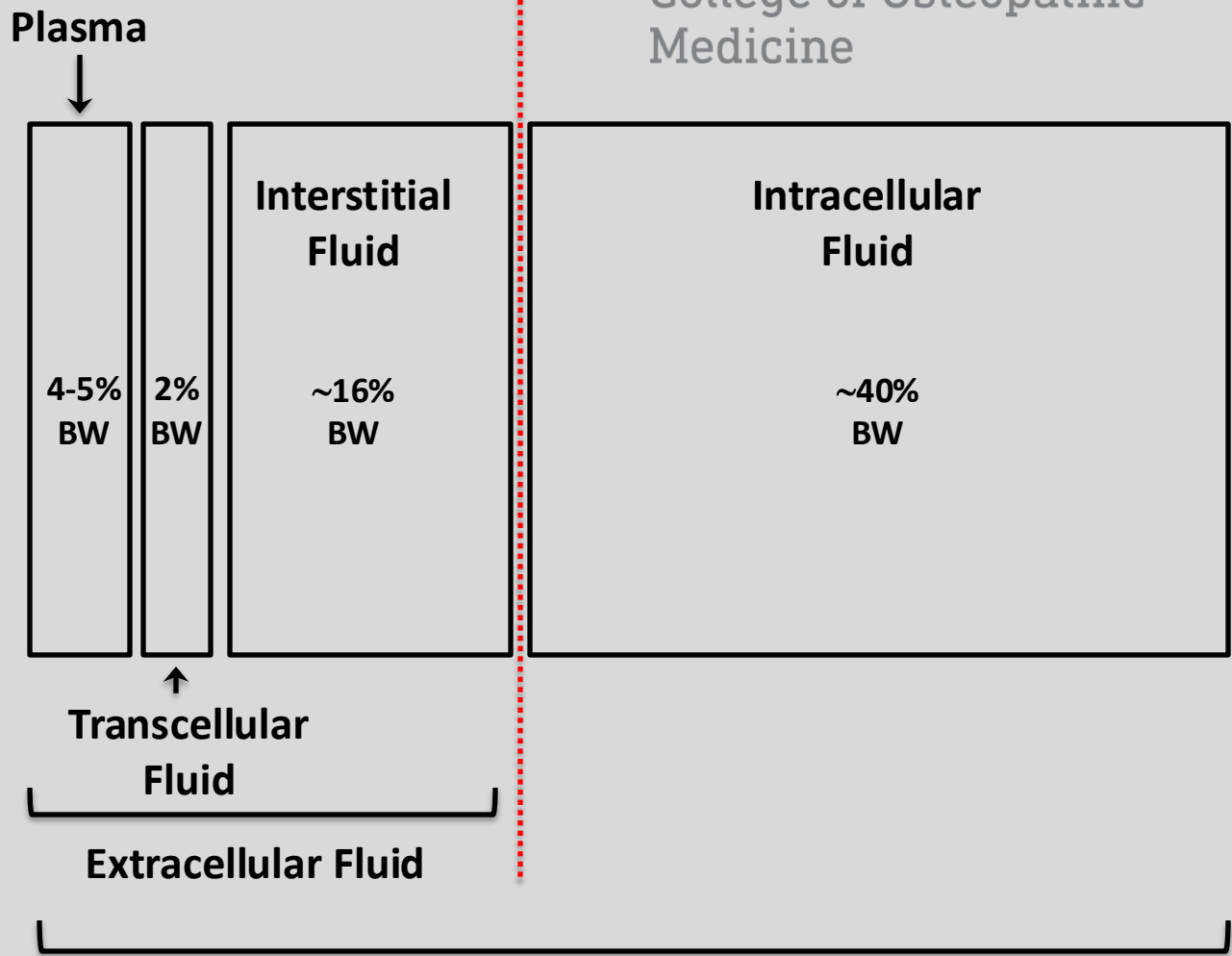
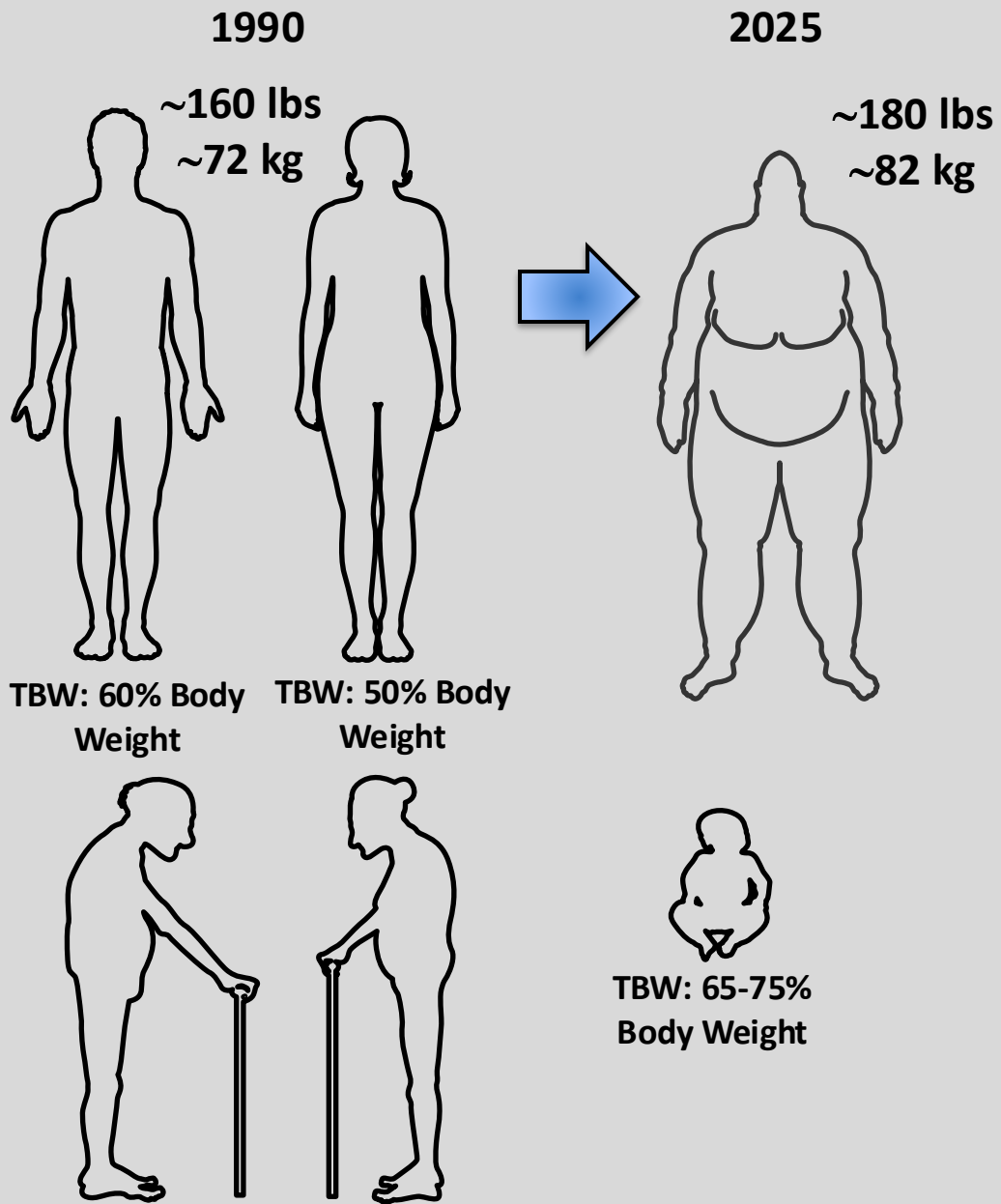
**Excessive
sweating**

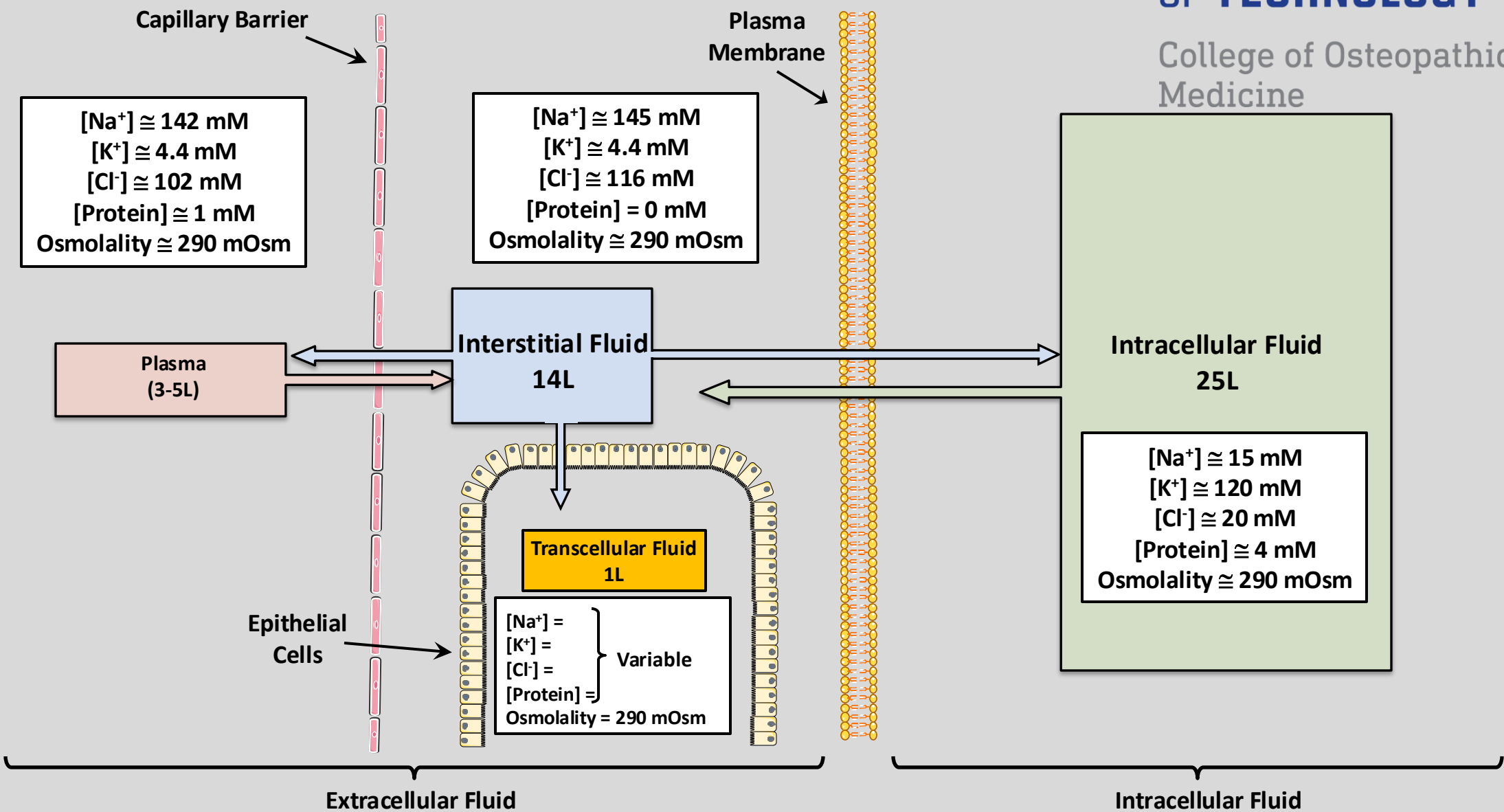
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**Prominent vasculature
Weight loss?**

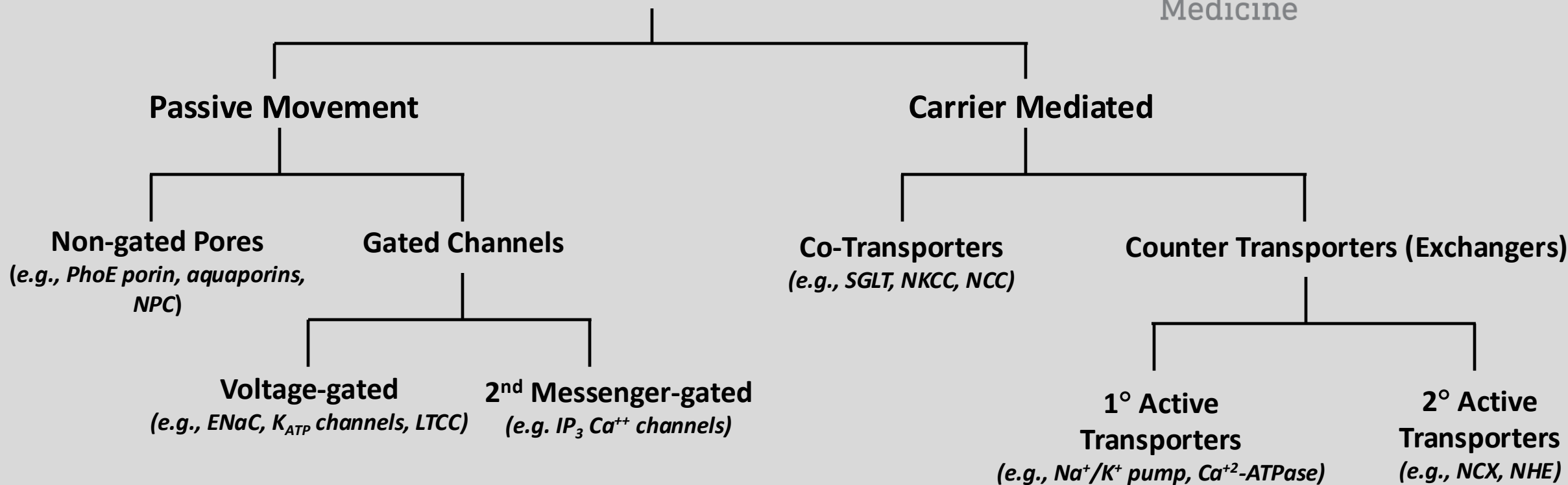
Fluid Compartments





Solute Transport

Movement Through Plasma Membrane

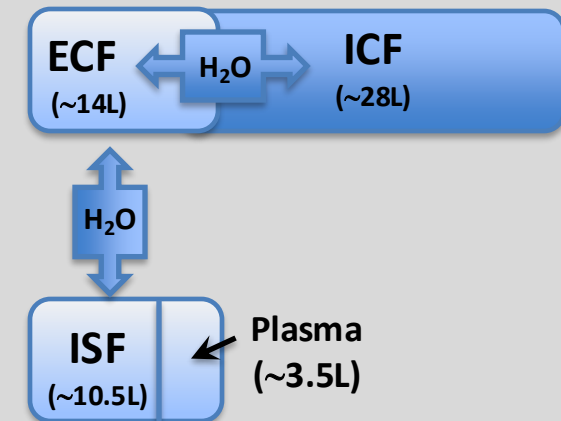
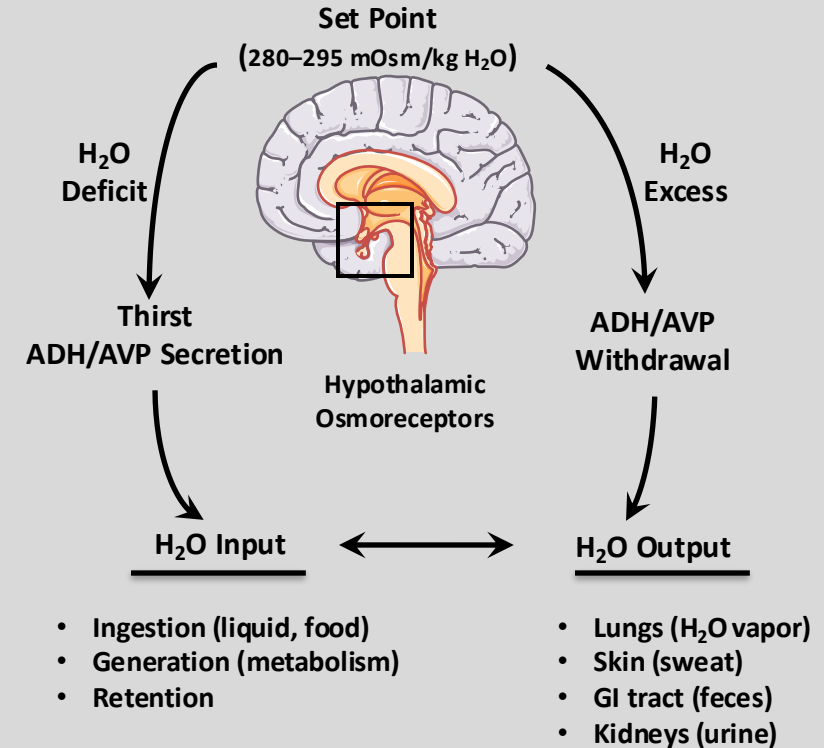


Movement of H₂O & Homeostasis

- Normal cell function requires the IC milieu (e.g., ions, small molecules, H₂O, H⁺) be maintained within a narrow range.
- Accomplished by the transport of substances and H₂O in & out of the cell.
- Additionally, nutrients are ingested & waste products are excreted
- In a healthy individual, these occur without significant changes in the volume or composition of the body fluid compartments.
- Maintenance of constant volume and composition of body fluid compartments is termed HOMEOSTASIS.
- The human body has multiple systems designed to achieve homeostasis.
- The human body is an “open system”.

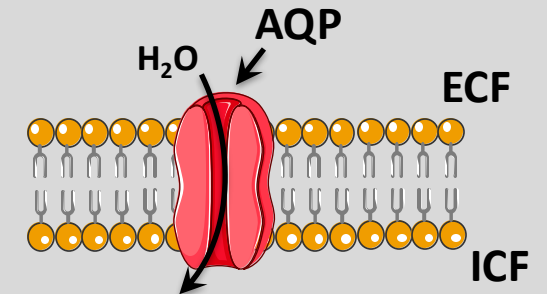
Substances gained $\cong$ substances lost $\rightarrow$ Steady-State Balance

- Concepts underlying steady-state balance:
 - Must have a “set point”
 - Must be able to generate effector signals that can lead to input/output
 - Effector organs must be able to respond
 - Must be sensitive to changes
- H₂O moves between the ICF and ECF across the plasma membranes
- H₂O moves between the vascular (plasma) and interstitial compartments across capillary walls.
- Pathways and driving forces for H₂O movement are different across cell membranes & capillary walls.



Movement of H₂O & Aquaporins

- H₂O transport across biological membranes is always passive.
- Single molecules can dissolve in lipid bilayers and move across cell membranes at a LOW but finite rate via simple diffusion.
- Diffusion through the lipid bilayer depends on its lipid composition.
- Movement of water across cell membranes occurs through transmembrane AQUAPORINS (passive conduits for H₂O).
- Large family of intrinsic membrane proteins
- Facilitate the rapid transport of water across cell membranes via pores
- AQP3, AQP7, AQP9, and AQP10 are also called aquaglyceroporins (permeable to both glycerol and H₂O)
- Driving force for this movement is the osmotic pressure difference ($\Delta\pi$)
- The osmotic pressure (π) of ICF & ECF are determined by ions (e.g. Na⁺, Cl⁻) & molecules (glucose, proteins).
- π is not a “pressure” in the sense of hydrostatic pressure.
- It just means osmolality.
- In reality (and numerically), it is just real osmolality expressed in pressure units (mmHg).
- The π of ICF & ECF ~280-295 mOsm/kg H₂O
- Cell volume remains constant when ECF osmolality is constant
- Therefore π inside the cells (π_{IC}) is equal to that outside (π_{EC}).
- If $\Delta\pi$ did exist, the cells would either swell or shrink



Isoform	Organ/Tissue
AQP0	Eye lens
AQP1	Brain, Blood vessels (BV), Eye, Kidney, RBC
AQP2	Collecting duct
AQP3	Eye, Immune system, Kidney, RBC, Skin
AQP4	Bronchi, Brain, Eye, Ear, Kidney, Tear glands, Muscle, BV
AQP5	Bronchi, Eye, Tear glands, Salivary glands, Sweat glands
AQP6	Kidney
AQP7	GI tract, Kidney, Liver
AQP8	GI tract, Kidney, Liver
AQP9	Liver, RBC, CV system, Muscles
AQP10	Small intestines
AQP11	Kidney, Liver, Testes
AQP12	Pancreas

Movement of H₂O & Hydrostatic Pressure

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- π is proportional to osmolality & T°
- This equivalence is expressed by the van't Hoff equation:

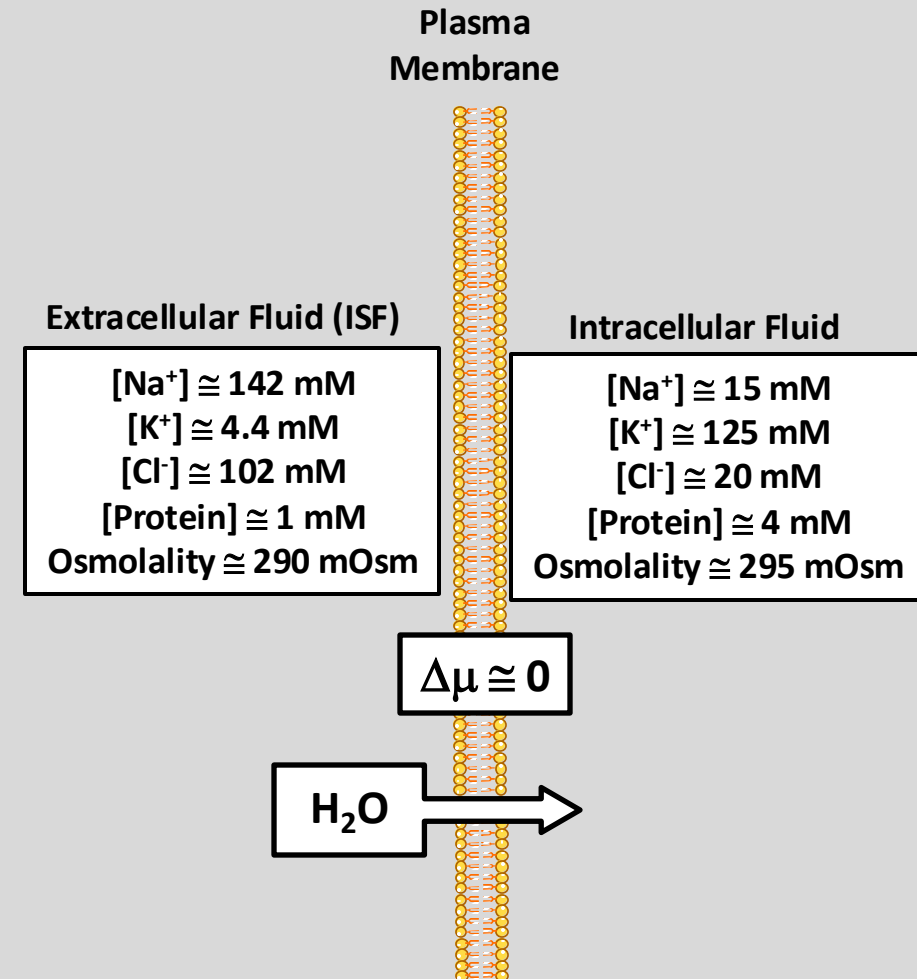
$$\pi = RTC$$

R = Gas constant

T = Absolute T°

C = Osmolality

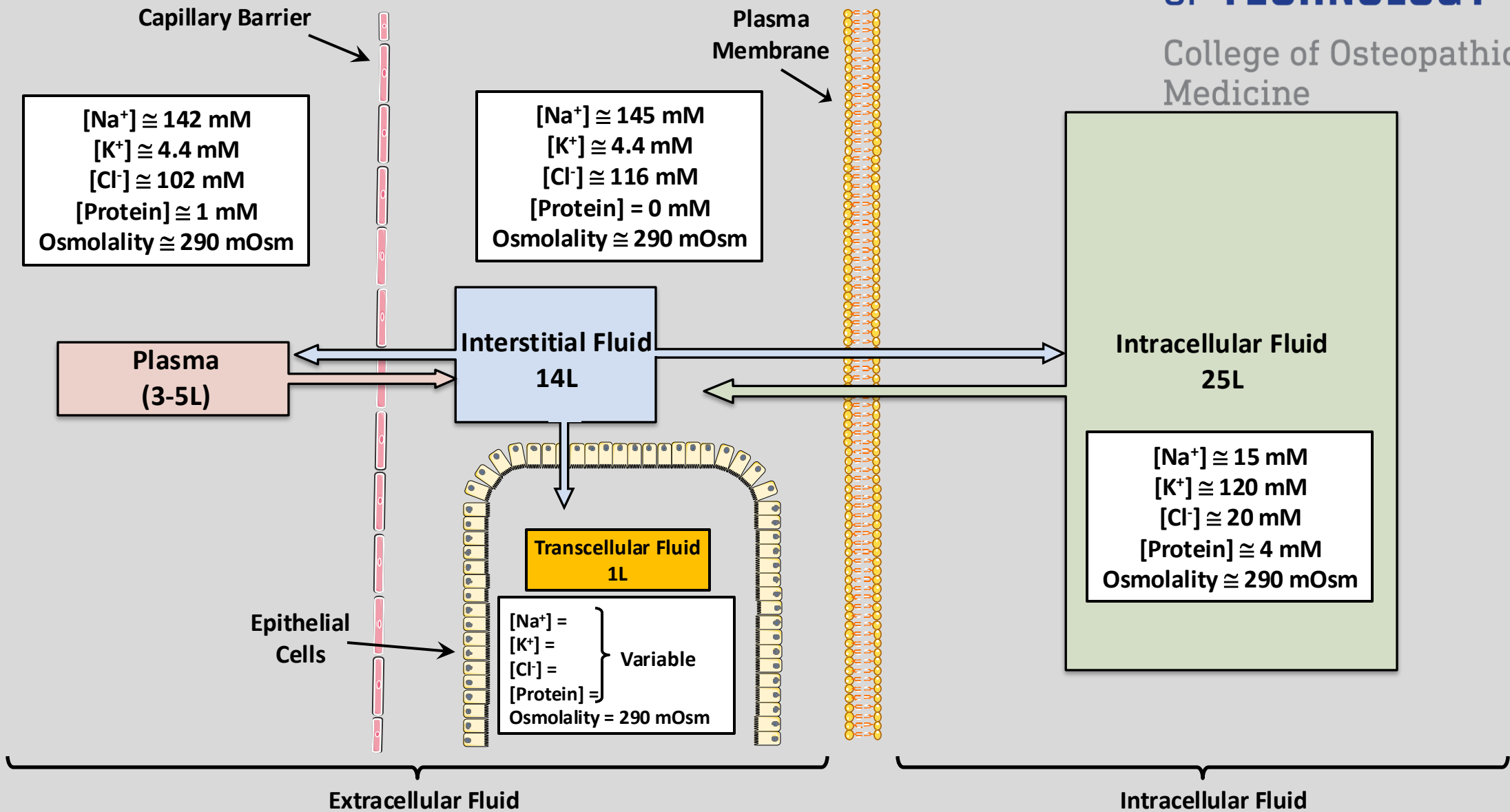
- Small differences in osmolality can lead to significant pressure differences (e.g., An $\Delta\pi$ of 1 mOsm = ΔP of 19.3 mmHg at 37°)
- H₂O moves by osmosis toward the solution with the higher osmolality.
- It is convenient to think of solutes as “pulling” water.
- Movement of H₂O (driven by osmotic gradient) = OSMOSIS
- Not the only driving force.
- The 2nd is the energy difference that results from the difference in HYDROSTATIC PRESSURE ($\Delta\mu$)
- H₂O movement is dependent on $\Delta\pi$ & $\Delta\mu$.
- Plasma membranes of animal cells are not rigid (unlike plant cells)
- Can't tolerate any significant $\Delta\mu$ without deformation.
- Therefore, the $\Delta\mu$ across a cell membrane is virtually always near zero
- Not a significant driving force for H₂O transport.



Summary (so far)

- Pathways and driving forces for H₂O & solute movement are different across cell membranes & barriers.
- Via simple diffusion, H₂O molecules can dissolve in the lipid bilayer and move across cell membranes at a very LOW but finite rate.
- Diffusion through the lipid bilayer depends on its lipid composition.
- Movement of water across cell membranes occurs mainly through transmembrane AQUAPORINS (H₂O channels).
- Hydrostatic pressure is NOT a significant driving force for CELLULAR (plasma membrane) transport of H₂O.
- Increasing the interstitial osmotic pressure, and thus generating a $\Delta\pi$, is quite a different matter.

Anatomy of Body Fluid Compartments

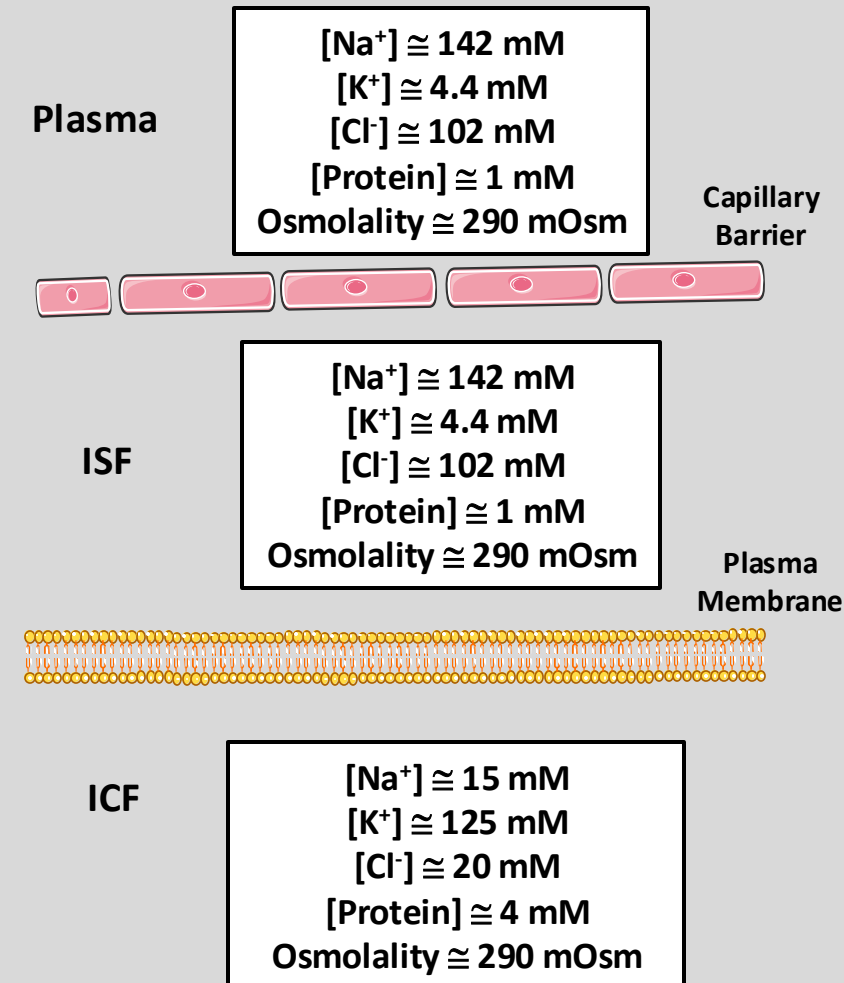


Proteins & Regulation of Cell Volume

- How do the different gradients of ALL the substances in the interstitial & intracellular milieu contribute to H₂O movement?
- The product of diffusible ions on one side of the membrane will be equal to the product of diffusible ions on the other side of the membrane.
- The electrochemical gradient produced by unequal distribution of charged ions creates a transmembrane potential difference (a.k.a. Membrane Potential), calculated using the Nernst equation.
- NaCl is largely excluded from the intracellular compartment by Na⁺/K⁺ ATPase.
- In the absence of Na⁺/K⁺ ATPase activity, cells will swell even when $IC_{Osm} = EC_{Osm}$
- This paradox is due to the presence of negatively charged, impermeable macromolecules (i.e., cytosolic proteins) on one side of the membrane
- This creates an osmotic diffusion gradient
- This attracts H₂O into that compartment.
- This is due to the GIBBS-DONNAN EQUILIBRIUM.
- Guides the behavior of charged particles near a semi-permeable membrane.

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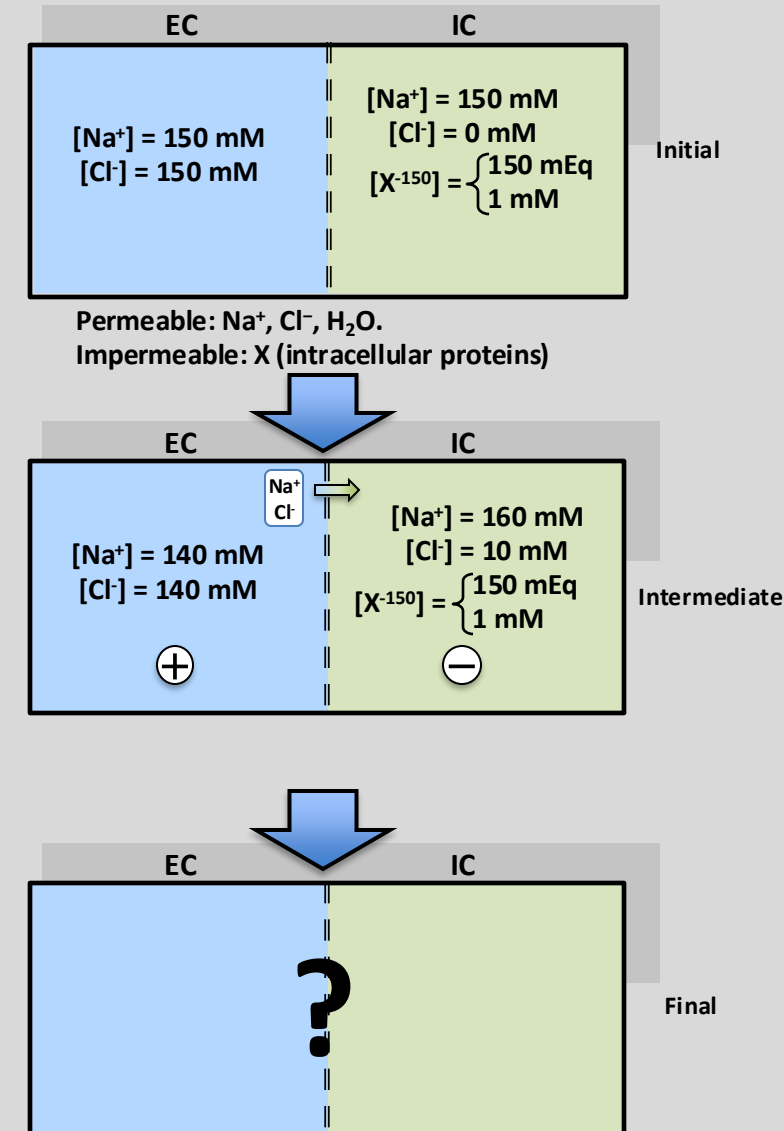


Proteins & Regulation of Cell Volume

Gibbs-Donnan Equilibrium:

- Determines how cells and blood vessels manage electrical charge, ion concentrations, and fluid movement.
- Named after Josiah Willard Gibbs (1839–1903), an American engineer and scientist & Frederick George Donnan (1870–1956), a British chemist.
- *Note: Most proteins are negatively charged in the physiological environment (pH ~ 7.2-7.4, due to high proportion of acidic amino acids (Glu, Asp) and PO_4^{3-})
- In a space separated by a semipermeable membrane, Cl^- will move down its concentration gradient into the cell.
- Total number of positive charges must balance the total number of negative charges (bulk electroneutrality) at all times.
- As $[\text{Na}^+]$ enters the cell $\rightarrow [\text{Na}^+]_{\text{IC}} \uparrow$ & $[\text{Na}^+]_{\text{EC}} \downarrow$
- This generates an inside-negative membrane voltage $\rightarrow$ attracts Na^+ , creating a membrane potential (V_m).

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Proteins & Regulation of Cell Volume

- As Na^+ enters the cell $\rightarrow \uparrow [\text{Na}^+]_{\text{IC}} \ \& \ \uparrow [\text{Cl}^-]_{\text{IC}}$
 $\downarrow [\text{Na}^+]_{\text{EC}} \ \& \ \downarrow [\text{Cl}^-]_{\text{EC}}$
- This condition is still far from equilibrium because the $[\text{Na}^+]_{\text{EC}}/[\text{Na}^+]_{\text{IC}}$ and $[\text{Cl}^-]_{\text{IC}}/[\text{Cl}^-]_{\text{EC}}$ ratios are not equal.

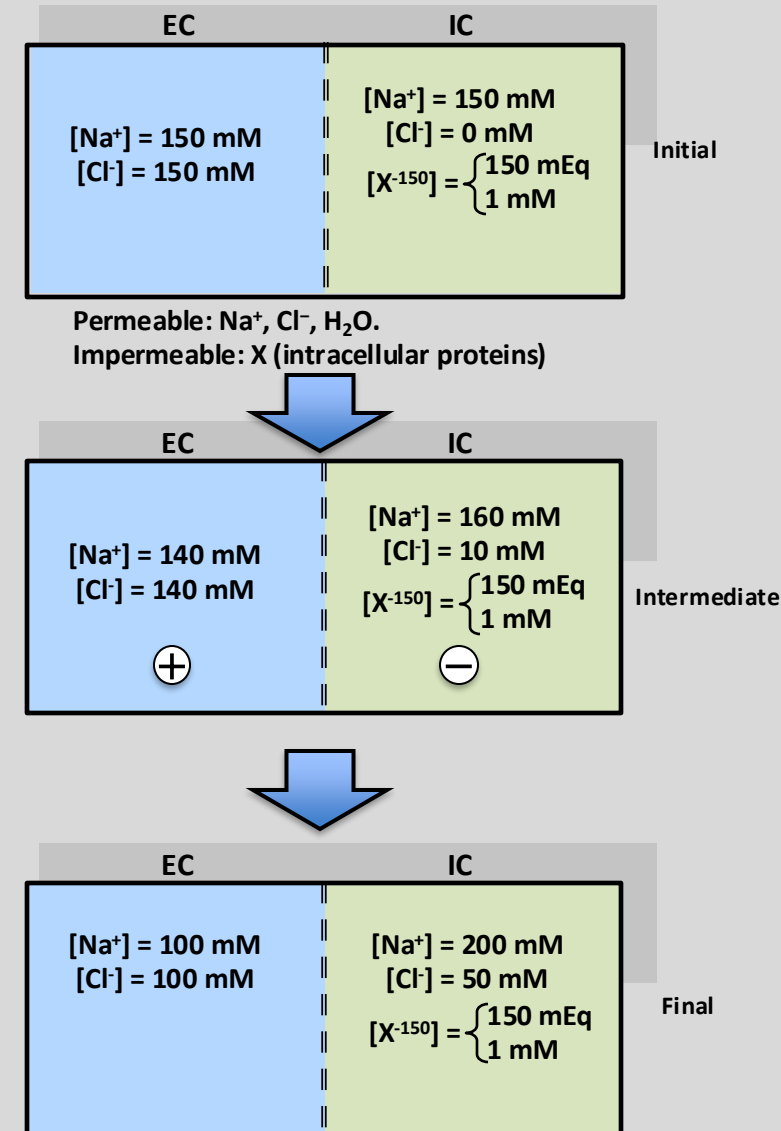
$$[\text{Na}^+]_{\text{EC}}/[\text{Na}^+]_{\text{IC}} \neq [\text{Cl}^-]_{\text{IC}}/[\text{Cl}^-]_{\text{EC}}$$

- Constraint on the system, as it approaches equilibrium, in each compartment
 Total # positive charges = Total # negative charges
- Therefore, Na^+ & Cl^- continue to move into the cell until Na^+/Cl^- are equal
- The final ratio corresponds to Nernst potentials of both ions.
- In the final equilibrium condition, $[\text{Na}^+]$ and $[\text{Cl}^-]$ will be distributed so that the concentration of each is balanced

$$[\text{Na}^+]_{\text{EC}}/[\text{Na}^+]_{\text{IC}} = [\text{Cl}^-]_{\text{IC}}/[\text{Cl}^-]_{\text{EC}} = \text{The Donnan Ratio } (r)$$

- Ions are in equilibrium but more osmotically active particles are now on the inside than on the outside.
- Ignoring $[\text{X}^{-150}]$, $[\text{Na}^+]_{\text{EC}} + [\text{Cl}^-]_{\text{EC}} > [\text{Na}^+]_{\text{IC}} + [\text{Cl}^-]_{\text{IC}}$
- Because of this gradient (ΔOsm), H_2O cannot be at equilibrium and will move into the cell.
- Only a minuscule amount of H_2O needs to enter the cell to generate a hydrostatic pressure to oppose the additional net entry of H_2O .
- This equilibrium hydrostatic pressure difference ($\Delta\mu$) opposes the osmotic pressure difference ($\Delta\pi$).

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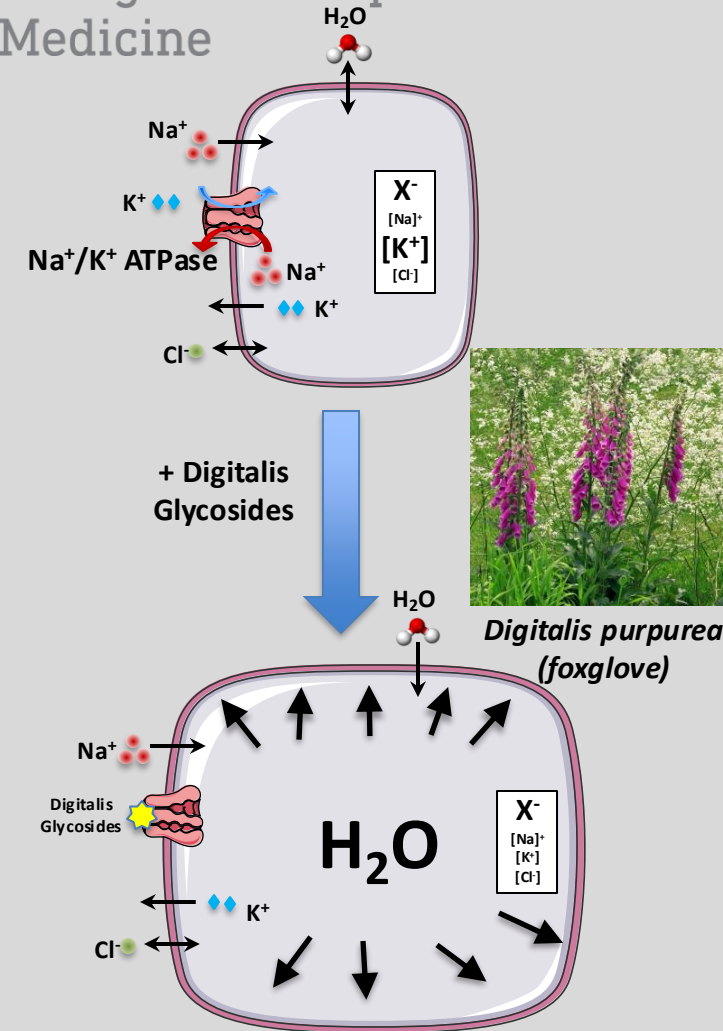


Na⁺/K⁺ ATPase & Regulation of Cell Volume

- Plasma membranes of mammalian cells are not rigid but deformable, so that transmembrane $\Delta\mu$ cannot exist.
- Distribution of ions toward the Donnan ratio leads to H₂O entry, cell swelling & bursting.
- The Donnan equilibrium model ignores all ions other than Na⁺, Cl⁻, and X⁻¹⁵⁰
- It illustrates an important point that the negative charge on impermeant intracellular solutes (e.g., proteins and PO₄³⁻) will lead to bursting.
- Cell must do “osmotic work” to counteract the passive swelling.
- Osmotic work: To exclude NaCl from the cell and make the cell functionally impermeable to NaCl.
- NaCl in the ISF/ECF offsets the osmotic effects of IC negative charges.
- A steady state maintained by active transport.
- Response to Cell Swelling:
 - $\downarrow ECF_{Osm} \rightarrow$ attracts H₂O into the cell.
 - Cells respond by activating solute efflux.
 - This response is known as a Regulatory Volume Decrease (RVD).
- Response to Cell Shrinkage:
 - $\uparrow ECF_{Osm} \rightarrow$ Draws H₂O out of the cell.
 - Cells respond by activating solute uptake.
 - This response is known as a Regulatory Volume Increase (RVI).

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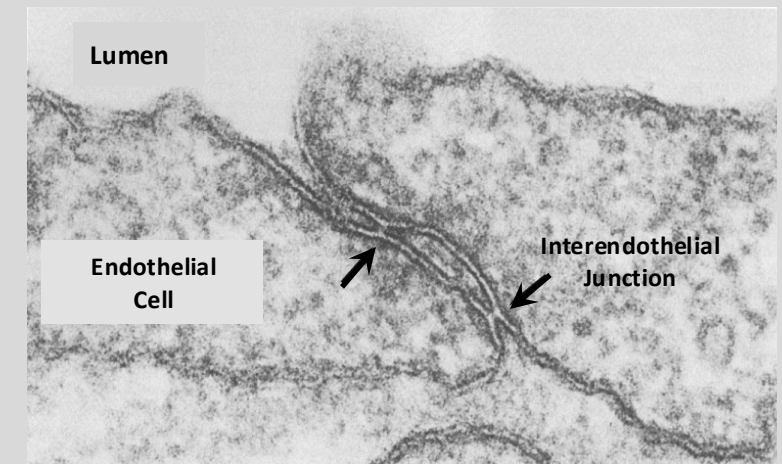
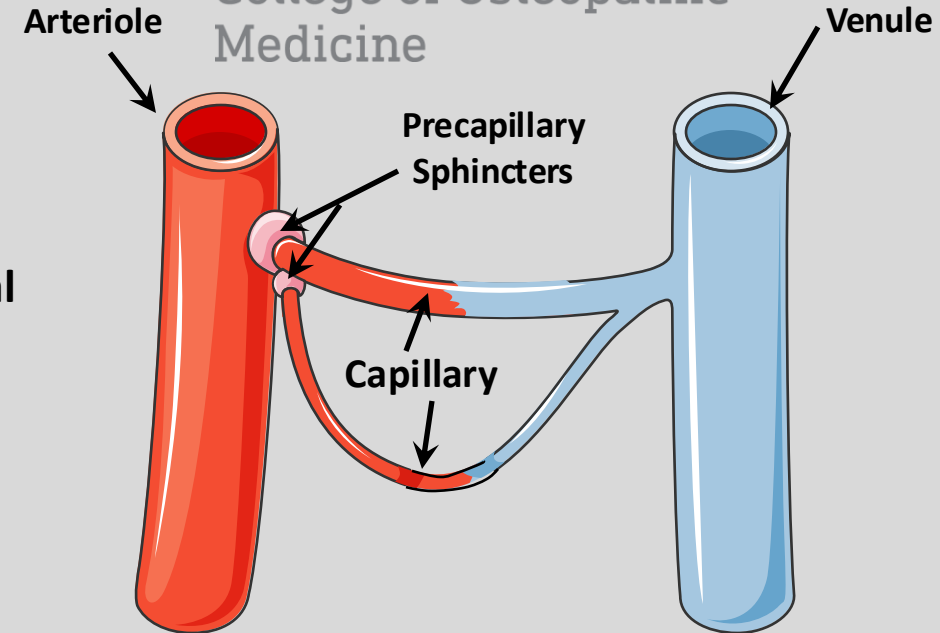


The Microcirculation

- 1° function of the cardiovascular system → Maintaining a suitable environment for tissue survival & function.
- The microcirculation is the “business end” of the system.
- Capillary:
 - Principal site for gas, H₂O, nutrient and waste-product exchange
 - Blood vessels from the 1st-order arterioles to the 1st-order venules
- In most tissues, flow exclusively serves nutritional needs.
- In some tissues, a large portion of capillary flow is non-nutritional (e.g., renal glomeruli, A/V anastomoses in skin)
- Precapillary sphincters:
 - Found at transition between a capillary and some arterioles.
 - Small cuff of smooth muscle cells
 - Usually not innervated
 - Responsive to local tissue conditions (paracrine & autocrine influences)
 - Function:
 - Control the access of flow to particular segments of the network
 - Relaxation/contraction of the precapillary sphincter modulates flow
- EC are linked together via interendothelial, adhering & tight junctions.
- Proteins such as claudins 1, 3, and 5 (CLDN1, CLDN3, CLDN54) as well as occludin (OCLN) are part of tight junctions.

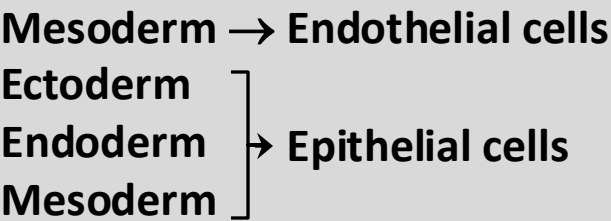
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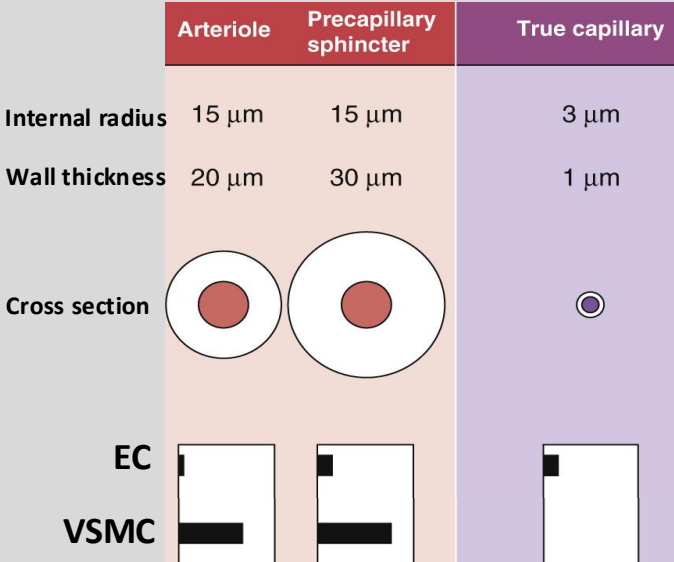


Epithelial & Endothelial Barriers

- Epithelium: An uninterrupted sheet of cells that are joined together by junctional complexes
- Vascular endothelium: Highly specialized, specific type of epithelial tissue.
- They have different embryological lineages:

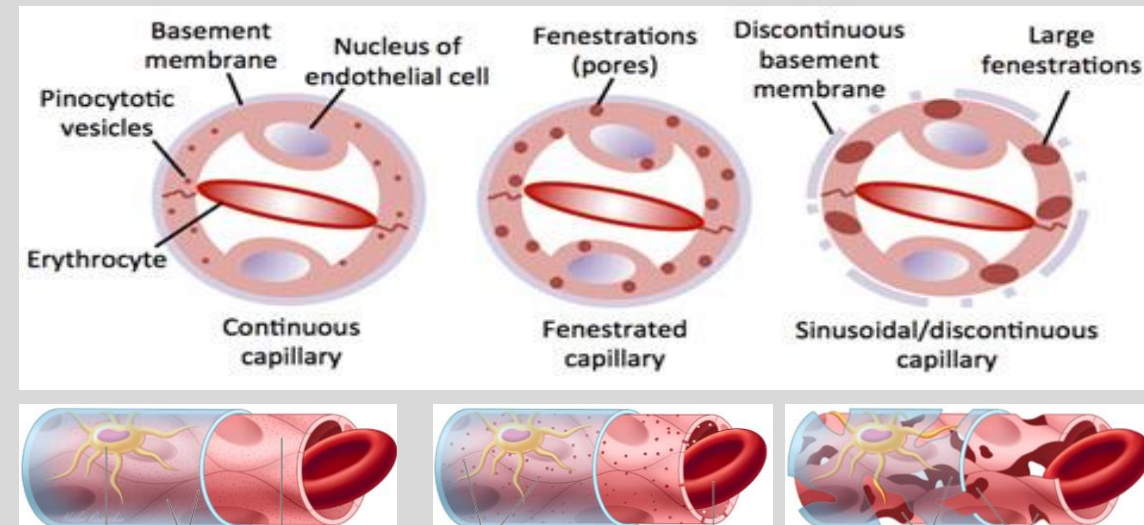
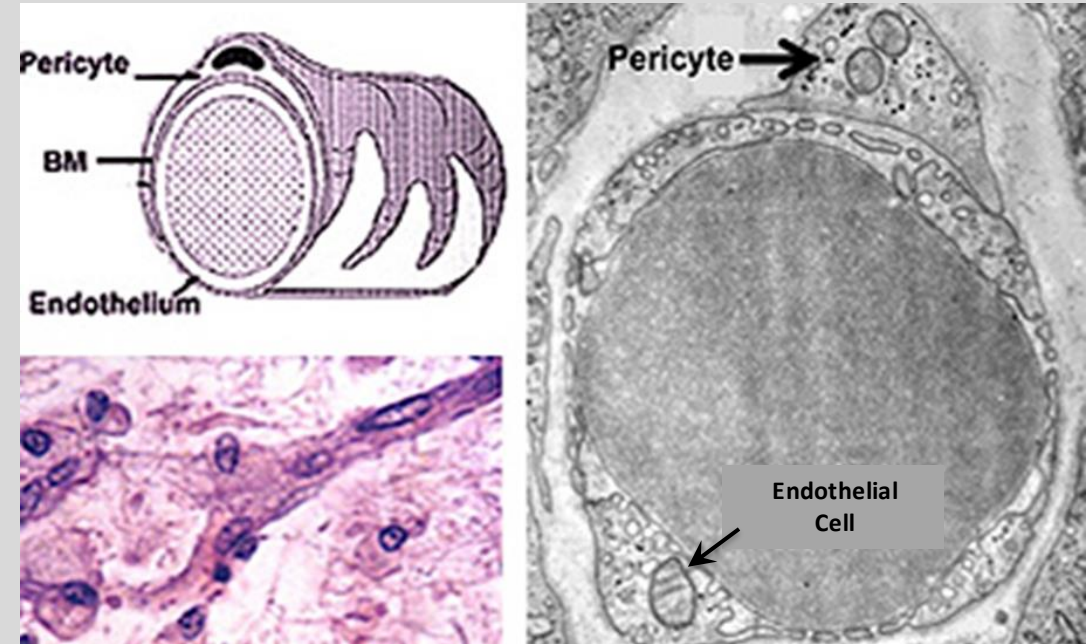


	Epithelium	Endothelium
Cell Type	Simple cuboidal & columnar Stratified squamous, cuboidal & columnar	Simple squamous
Location	Internal cavities & external surfaces, luminal surfaces	Internal surface of the CV & lymphatic systems, capillaries
Surface	Papillary projections (brush borders)	Smooth
Polarity	Yes (basolateral and apical poles)	No
Secretions	Mucous, PGE ₂ (gastric, pulmonary)	Nitric oxide, PGI ₂ (prostacyclin), growth factors, vasoactive proteins (vascular)
Regeneration	Rapid	Limited

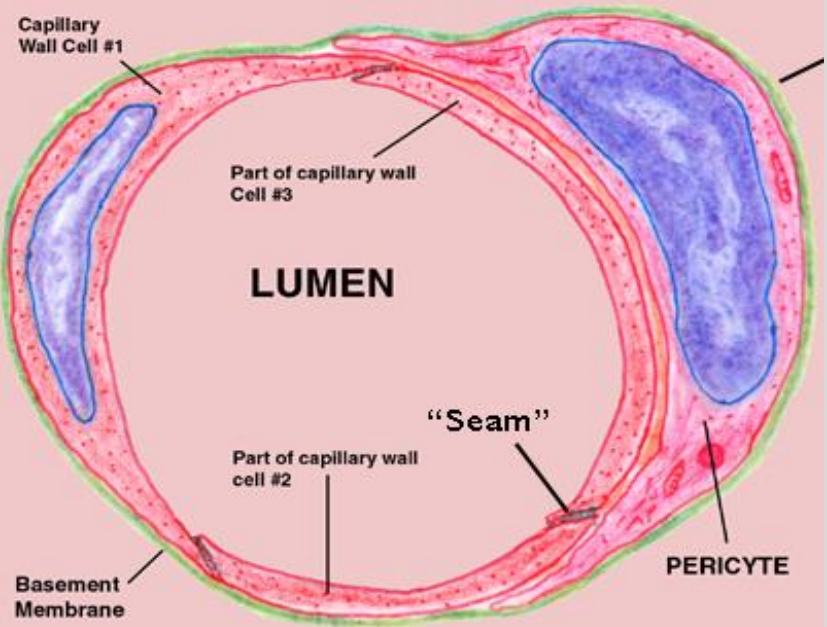


Capillaries

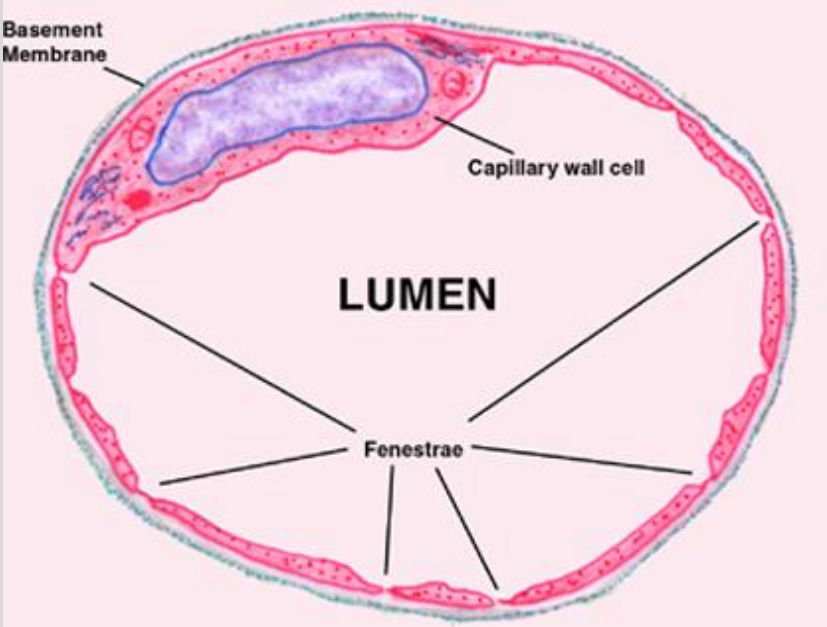
- A single continuous layer of EC's surrounded by a basement membrane and a network of reticular collagen fibers & interspersed pericytes.
- Pericytes:
 - Also known as mural cells.
 - Spatially isolated contractile cells (NOT SMC).
 - Essential for the regulation of blood flow and maintenance of homeostasis.
- Capillaries are categorized into 3 groups (based on their degree of "leakiness.")
 - Continuous capillaries:
 - Most common form of capillary with interendothelial, tight & adhering junctions (e.g., blood-brain barrier).
 - Fenestrated capillaries:
 - EC's are thin & perforated with fenestrations (windows).
 - Most often surround epithelia (e.g., small intestine, exocrine glands, kidney).
 - Discontinuous (sinusoidal) capillaries:
 - In addition to fenestrae, possess large gaps.
 - Close contact between blood/tissue & exchange of materials (e.g., liver, spleen).



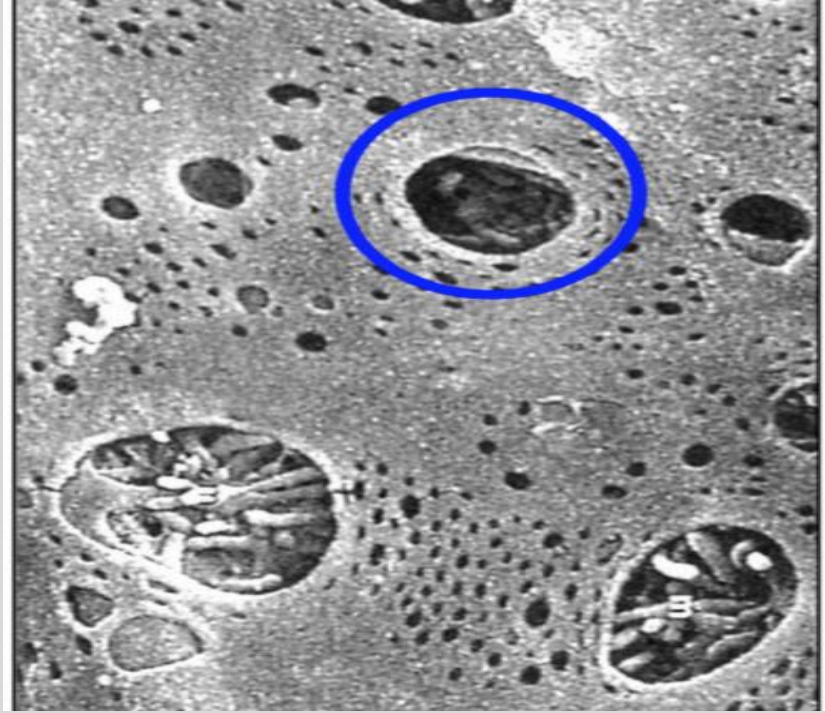
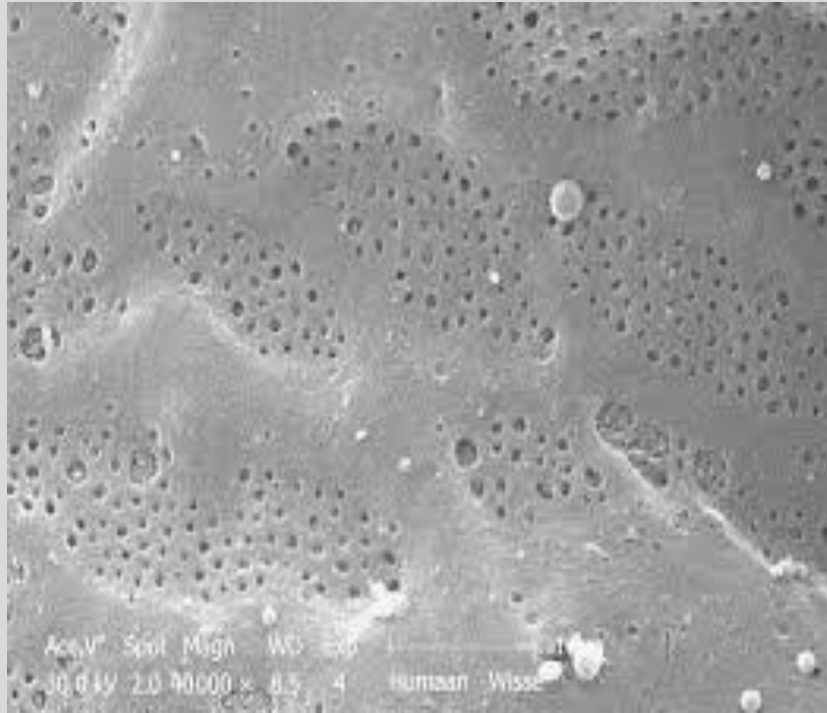
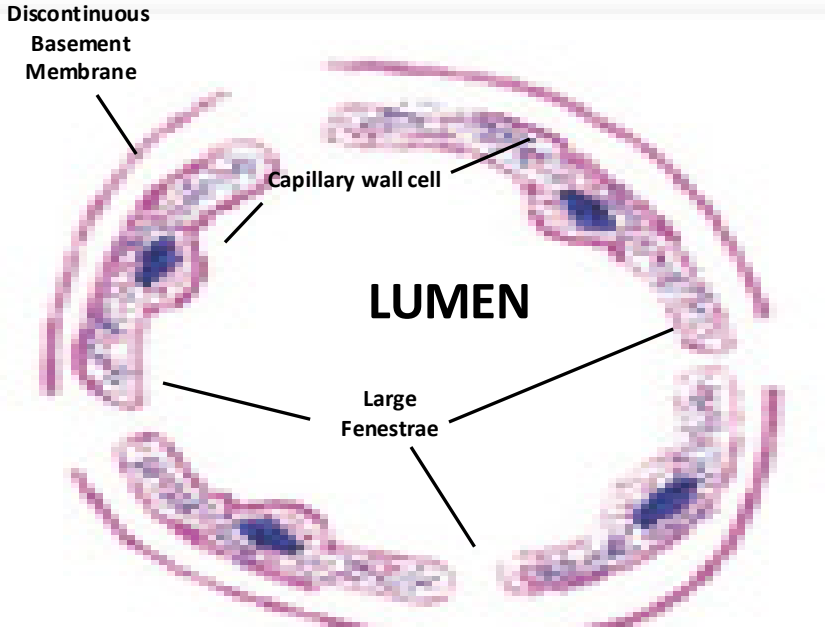
Continuous Capillaries



Fenestrated Capillaries



Discontinuous (Sinusoidal) Capillaries

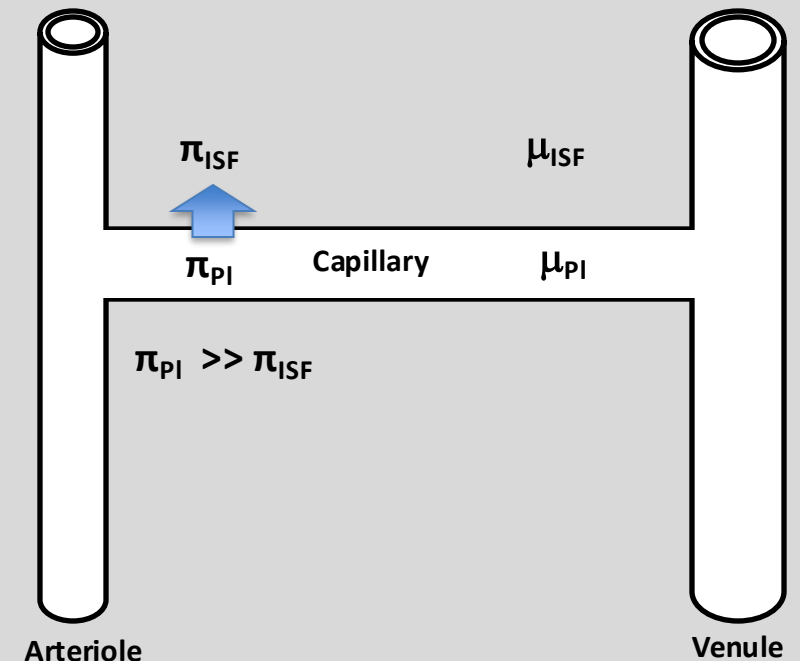
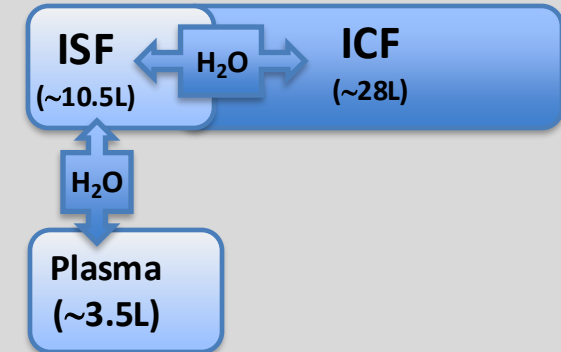


H₂O Exchange Across Capillary Walls

- Interstitial Fluid (ISF) surrounds and bathes all cells of the body (non-blood).
- ~75% (~14 L) of the ECF.
- Capillary walls separate the intravascular and interstitial compartments.
- In some tissues, a large portion of capillary flow is non-nutritional.
- Hydrostatic Pressure (μ):
 - Includes all sources of intravascular pressure
 - Not only derived from gravity
 - Used (mostly) in opposition to osmotic pressure.
- $\Delta\mu$ is an important force for driving fluid across cell barriers.
- Small solutes (e.g., electrolytes) permeate freely across most cell barriers.
- $\Delta\pi$ as a result of these small solutes DO NOT exert a driving force for H₂O flow across the cell barriers.
- The only net osmotic force that acts across the capillary wall is caused by the asymmetric distribution of plasma proteins in the ISF.

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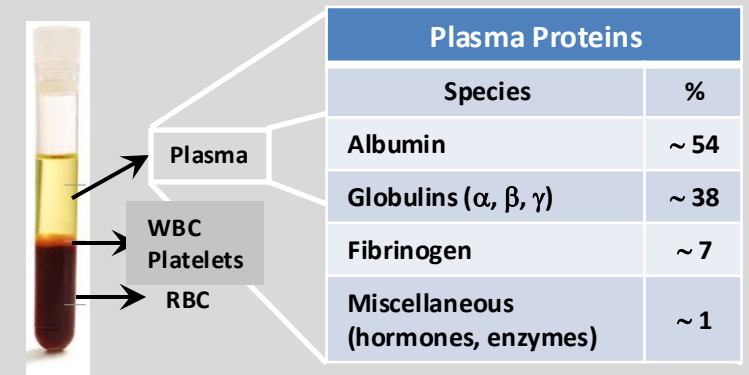


Solute Exchange Across Cellular Barriers

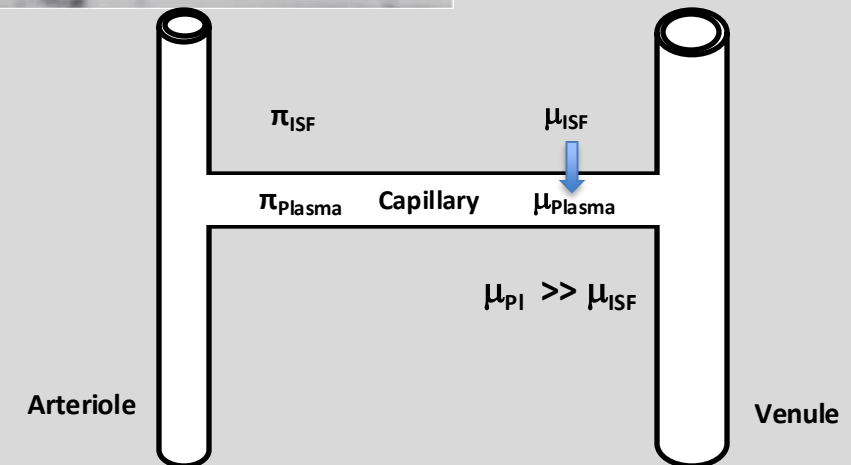
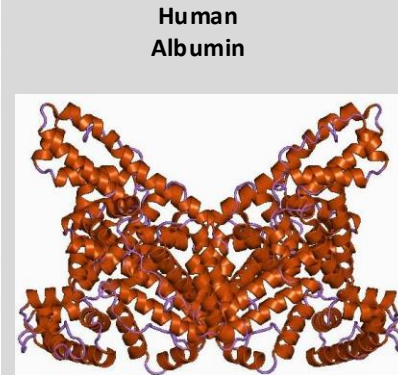
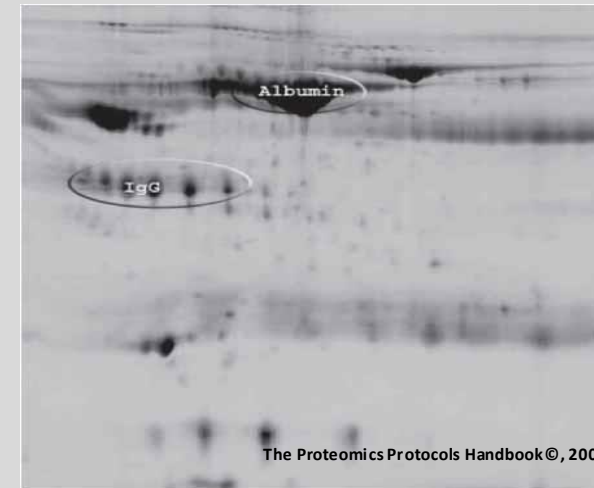
- Plasma proteins are too large to penetrate the capillary wall freely.

$$[\text{Protein}]_{\text{Plasma}} \gg [\text{Protein}]_{\text{Interstitial}}$$

- Higher $[\text{Protein}]_{\text{Plasma}}$ than $[\text{Protein}]_{\text{Interstitial}}$ sets up a $\Delta\pi$
- Pulls fluid back into the capillary.
- Called the COLLOID OSMOTIC PRESSURE (a.k.a. ONCOTIC PRESSURE, $\Delta\mu_{\text{onc}}$)
- ULTRAFILTRATION:** When the $\Delta\pi \gg \Delta\mu \rightarrow$ Movement of H_2O out of the capillary
- Presence of a greater concentration of plasma proteins in the intravascular compartment than in interstitial fluid $\rightarrow$ Pulls fluid back into the capillary.
- H_2O is at equilibrium across capillary wall when the colloid osmotic and hydrostatic pressure differences are equal.



2D Gel Electrophoresis-Human Plasma



Capillary Filtration Coefficient

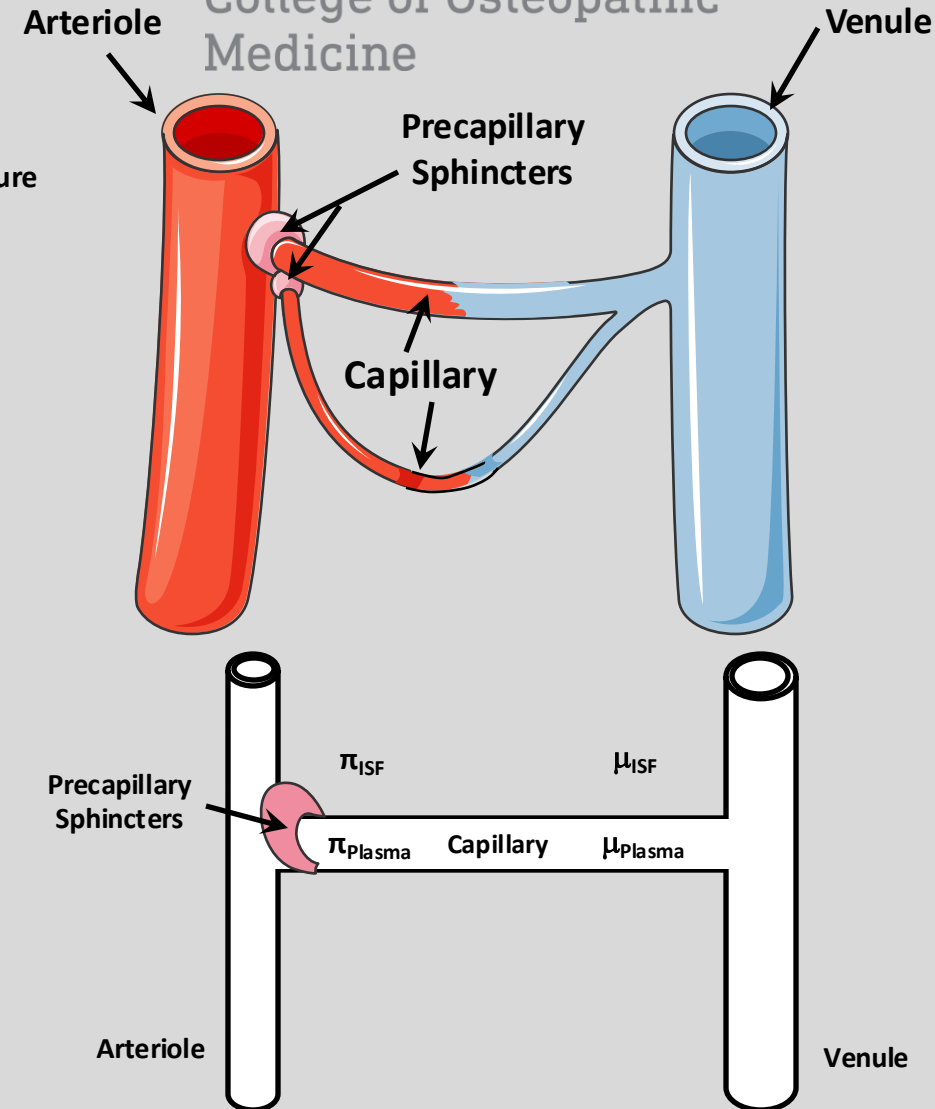
- The relationship between μ & π and their role in regulating fluid exchange across capillaries was expounded by Ernest Henry Starling (1866-1927).
- This relationship constitutes Starling's Hypothesis.

$$Q_f = k [(\mu_{PI} - \mu_{ISF}) - (\pi_{PI} - \pi_{ISF})]$$

- Filtration occurs when the result is positive
- Absorption occurs when it is negative.

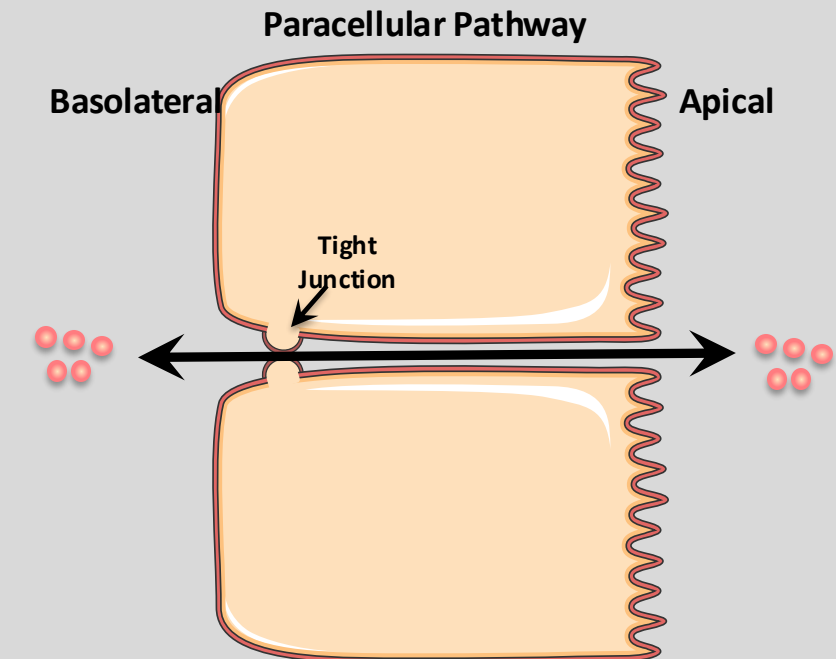
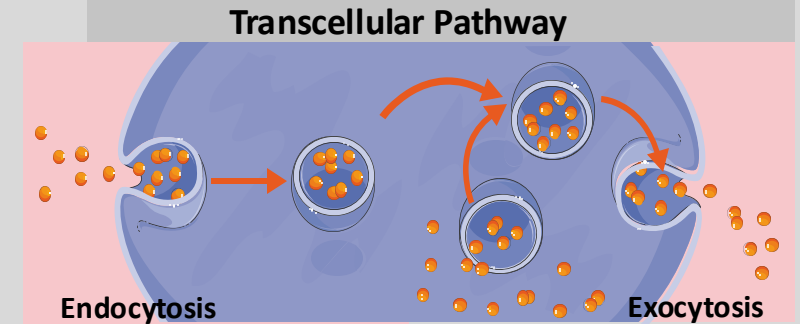
Q_f : Fluid movement
 k : filtration coefficient
 μ_{PI} : Capillary hydrostatic pressure
 μ_{ISF} : ISF hydrostatic pressure
 π_{PI} : Plasma oncotic pressure
 π_{ISF} : ISF oncotic pressure

- Capillary filtration: Volume of fluid filtered from the vascular to interstitial spaces.
- It is directly proportional to:
 - Capillary wall permeability
 - Total surface area (SA)
- Capillary filtration coefficient (k) is a quantitative measure of these two parameters.
- There are extreme differences in permeabilities and SA of capillaries in different tissues.
- Filtration/absorption can be adjusted based on these variables
- Precapillary sphincters $\rightarrow$ Capillary hydrostatic pressure
- SA can change depending on the tissue and the presence of contractile cells (e.g. mesangial cells, pericytes)



Solute Exchange Across Cellular Barriers

- Endothelial/epithelial cell membranes are freely permeable to gases and lipids.
- Capillary wall is a significant barrier to the exchange of lipid-insoluble substances.
- Negative charges in the endothelial glycocalyx exclude macromolecules with negative charge & favor macromolecules with positive charge.
- Flux of negatively charged proteins is less than that of neutral macromolecules of equivalent size.
- Transcellular Pathway: Translocation of molecules across a cell
 - Translocation of large molecules ($r > 1\text{nm}$) across the two cell membranes and cytosol (a.k.a. large-pore effect) can occur by endocytosis.
 - Endocytosis occurs by three mechanisms:
 - Pinocytosis: Nonspecific uptake of small molecules & H_2O
 - Phagocytosis: Cellular internalization of large particles (e.g., bacteria, cell debris)
 - Receptor-mediated endocytosis: uptake of specific molecules into the cell by molecules binding to cell surface receptors.
- Paracellular Pathway: Hydrophilic solutes (smaller than albumin) can bypass the cell and traverse the endothelial/epithelial wall by diffusion via clefts, interendothelial junctions & fenestrae.
- Tight and leaky epithelia differ in the permeabilities of their tight junctions.

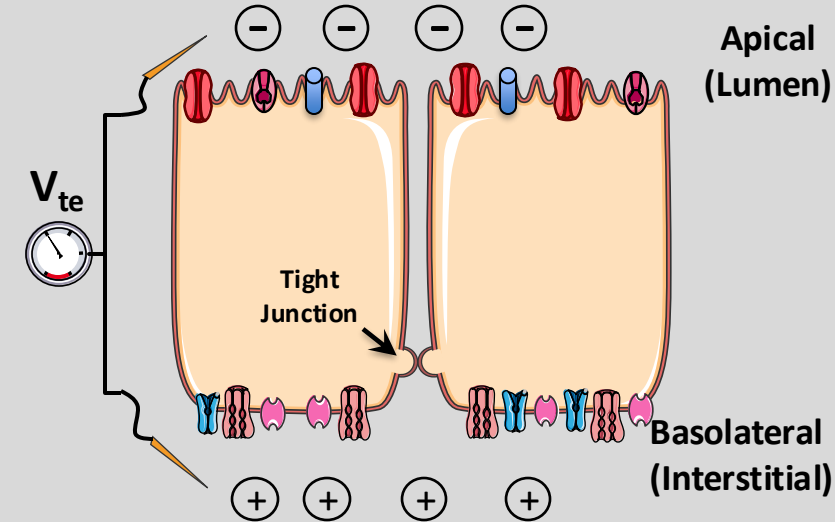


Electrochemical Gradient across Epithelial Barriers

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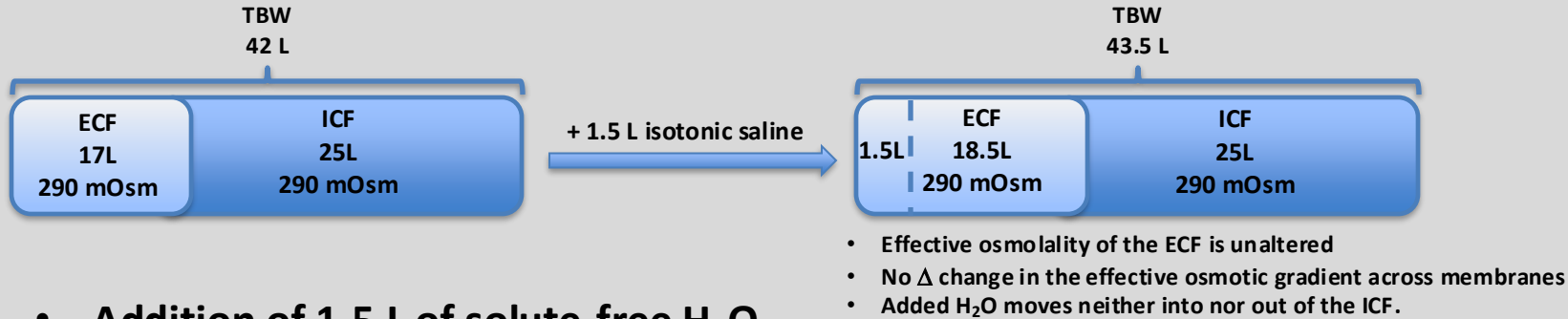
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- The composition of luminal fluid is not the same as that of ECF
- Transepithelial concentration differences occur as a result of the polarity of solute transport and transporters.
- The electrochemical gradients across the apical and basolateral membranes of epithelial cells are different.
- Transepithelial voltage (V_{te}): Voltage difference ($\Delta_{\text{potential energy}}$) between either side of the epithelium.
- One measure of how tightly an epithelium separates one compartment from another is its resistance to the flow of electrical current.
- High electrical resistance: Epithelial barriers with “tight” links
- Low electrical resistance: Epithelial barriers with “leaky” links
- In general:
 - Leaky epithelia perform bulk transepithelial transport of solutes and H_2O in a nearly isosmotic fashion (e.g., renal proximal tubule, small intestine)
 - Tight epithelia generate large transepithelial ion concentration/osmotic gradients (e.g., renal distal nephron, large intestine, urinary bladder)



Effects of Salinity on Body Fluid Compartments

- Adding combinations of NaCl and solute-free water to the ECF will alter the volume and composition of the body fluid compartments.
- Effects seen with applications of intravenous therapy.
- Addition of 1.5 L of isotonic (normal) saline (0.9% NaCl in H₂O)



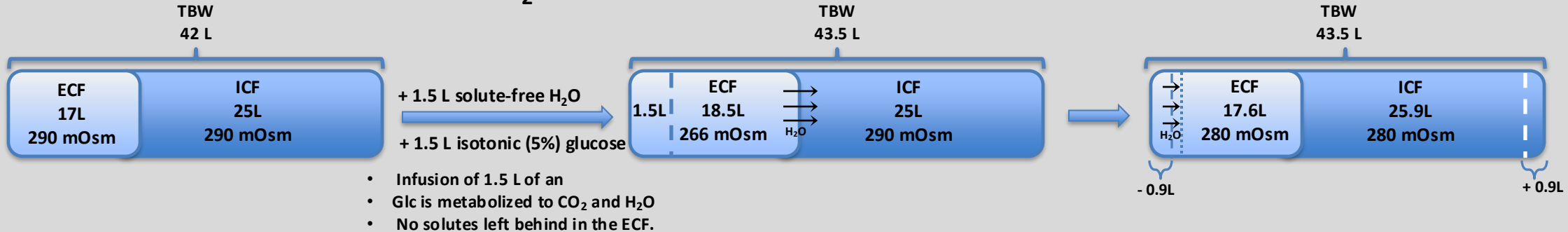
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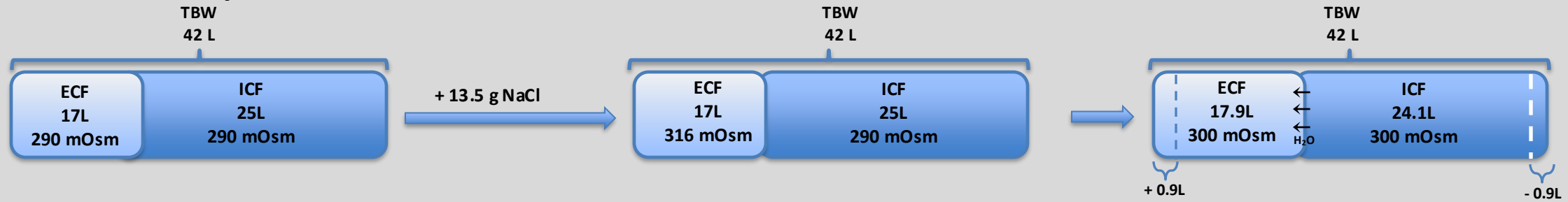
Body Fluids (70 kg person)

Compartment	%	Volume (L)
TBW	60 (of TW)	42
ICF	60 (of TBW)	25
ECF	40 (of TBW)	17

- Addition of 1.5 L of solute-free H₂O



- Addition of pure NaCl



Movement of H₂O Between Compartments

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Evidence for continuity of interstitial spaces across tissue and organ boundaries in humans

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